

Data Cleaning & Preparation Log

Dataset Overview

- Initial Records: 15,121
- Dataset Type: Aviation Operations, Fuel Consumption, and Emissions Data
- Objective: Prepare data for fuel-efficiency, emissions, route performance, and sustainability reporting analysis.

Data Quality Assessment

Issue Type	Records Affected	Action Taken	Business Impact
Duplicate Flight IDs	24	Removed duplicate records based on the unique flight identifier (<code>id</code>) after validating sample duplicates	Prevented double-counting of fuel burn, emissions, passenger traffic, and flight activity
Negative Distance Values	18	Removed records with negative Actual Distance KM values	Negative flight distances are physically impossible and would distort efficiency metrics
Zero Distance Values	13	Removed records with zero Actual Distance KM	Zero-distance flights cannot be used for fuel-efficiency and emissions intensity calculations
Passenger Capacity Violations	0	No action required	No records exceeded aircraft passenger capacity
Missing Fuel Data	3,708	Retained as null and flagged for analysis limitations	Missing fuel data impacts fuel-efficiency and emissions calculations
Block Time Issues	0	No action required	No invalid block time values identified
Flight Time Issues	0	No action required	No invalid flight time values identified

Missing Data Review

Columns with extremely high missingness were reviewed individually rather than automatically removed. Business-critical fields were retained even when missing values were present.

Actions taken:

- Removed columns containing 100% null values.
- Retained operational and emissions-related columns with missing values for transparency.
- No missing values were replaced with zero without business justification.
- Missing values were treated as unavailable information rather than actual operational values.

Data Type Standardization

The following data type corrections were performed to support accurate calculations and reporting:

Data Category	Data Type Applied
Flight Dates	Date
Distance Metrics	Decimal Number
Fuel Consumption Metrics	Decimal Number
CO ₂ Emissions Metrics	Decimal Number
Passenger Counts	Whole Number
Aircraft Capacity Fields	Whole Number
Block Time & Flight Time Metrics	Whole Number
Flight Identifiers and Codes	Text
Aircraft Type and Route Fields	Text

Assumptions

1. Duplicate IDs represented duplicated records after sample validation.
2. Negative flight distances were treated as invalid data entries.
3. Zero-distance flights were excluded from efficiency calculations.
4. Missing fuel values were not imputed due to insufficient business justification.
5. Blank values were retained as missing information rather than replaced with zero.
6. Columns with no populated values were removed from the analytical dataset.

Result

The cleaned dataset was prepared for:

- Fuel burn analysis
- CO₂ emissions analysis
- Aircraft efficiency comparison
- Route-level emissions analysis
- Passenger load factor analysis
- ETS and CORSIA reporting review